



Fundamentals of the Data-Driven Enterprise



L5 Data Engineer Higher Apprenticeship
Module 1 / 12 (“Data Fundamentals”)
Topic 1 / 4

rev. 1 (2024)

Module Overview

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(Fundamentals of
the) Data-driven
enterprise

Introduction to
data quality

Introduction to
managing data
projects and
products

Introduction to
reliable data
architectures



Webinar agenda

This webinar will cover the following:

- Settling into your new programme
- Building a data-driven culture
- Fundamentals of data
- Standards and engineering best practices

Webinar length: 3 hours

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Introductions

We are very excited to have you starting your journey with Data Engineering!

You have already introduced yourselves during the Induction, but let's continue getting to know one another!

- Intros and catch-up (2 minutes)
- What exciting data projects or products have you worked on (or would like to work on)? (3 minutes)
- Your two Goals and Expectations from this programme (2 minutes)
- How are you managing your time for e-learning? (3 minutes)

Share your experiences!

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Welcome to the programme!



Time management

Your tutor will now re-cap the main points that you discussed re: time management.

Do you want to add anything else at this point, or ask any more questions about the time expectations for this course?

Think about:

- are you a morning learner or an evening learner?
- do you need quiet focus or more stimulation?
- how do you communicate your time and space needs at work?
- do you want to track extra learning if you're aiming for a distinction?

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Hub, Programme Handbook and Tracking

Your tutor will remind you how to access the Hub, and will show you your forum, and your Skills Scan (I).

There will be a post on the forum with your submission deadlines for your programme's formatives and portfolio pieces.

Your tutor will also remind you how to access the Programme Handbook via the Hub, and you will explore some sections of it together.

Pay attention to how you can track extra learning activities such as reading, shadowing, internal research, and stretch/extend activities.

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KSBs

You will now review the KSBs for this programme.

These are in your Programme Handbook.

After you come back to the main room, discuss which KSBs did you find the most surprising, or perhaps even unclear at this point?

Remember you are at the very beginning of your journey, and things will start making more sense very soon!

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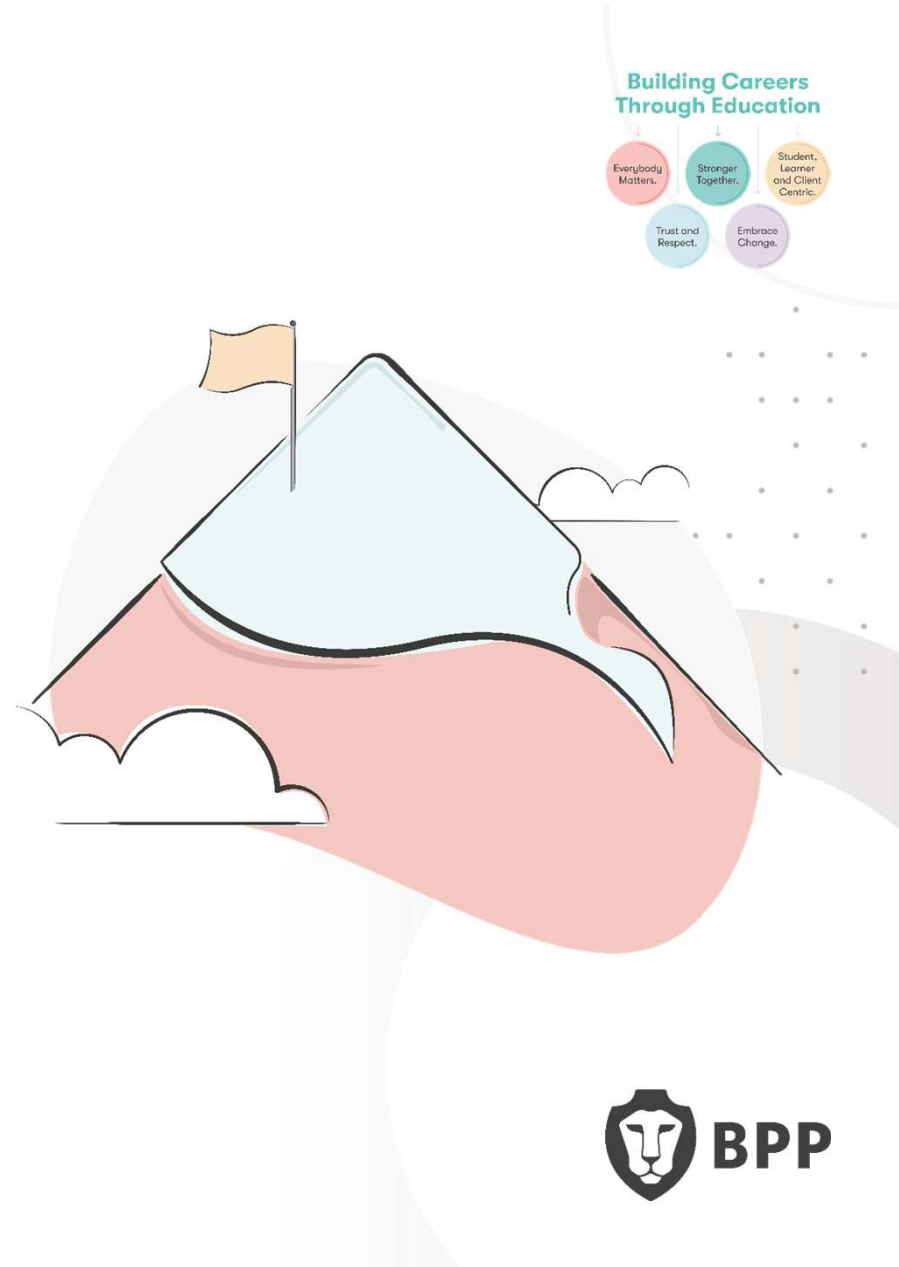


Learning objectives

By the end of today's webinar, you will be able to:

- Understand the **value** of data in modern organisations
- Become familiar with different **types** and **sources** of data
- Appreciate the significance of standards, best practices and **regulations**

Sounds like a lot? Don't worry! We will provide real-world examples for each of the key concepts that you learn about today.



Data-Driven Enterprises

A real-world success story...

- Netflix leverages data analytics for user experience, content recommendations, and streaming optimisation
- Through analysis of user behavior and preferences, Netflix tailors content and predicts successful shows
- This data-driven approach fuels subscriber growth, cementing Netflix's dominance in streaming



Netflix: A Highly Successful Data-Driven Organisation



The Vital Role of Data Engineering

Data engineering in action...



Harnessing the power of big data



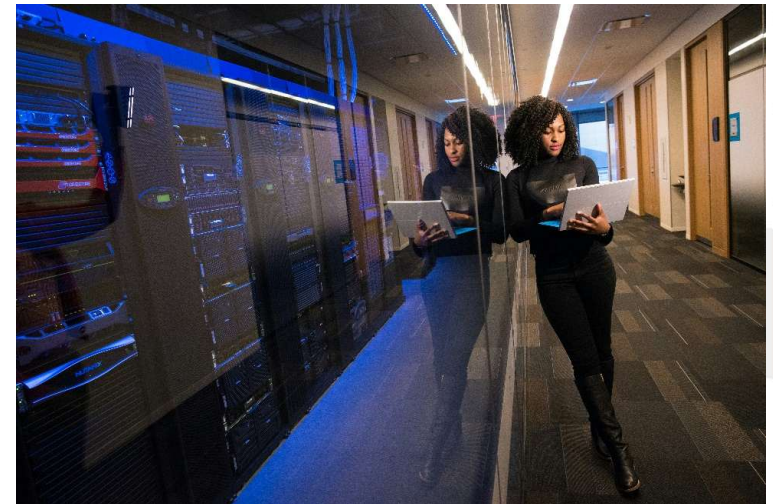
Collecting, storing, and processing data at scale



Integrating data sources, and ensuring data quality



Driving innovation and growth



The role of data engineering and data engineers

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Understanding Poll

The e-learning for this week's topic covered a wide range of concepts.

So, are there any concepts you would like a further explanation or support with?

Concepts covered included:

- What is a data-driven culture?

Your tutor will now **whiteboard** with you the elements of a data-driven culture, such as:

Clean, usable data

Governance and quality

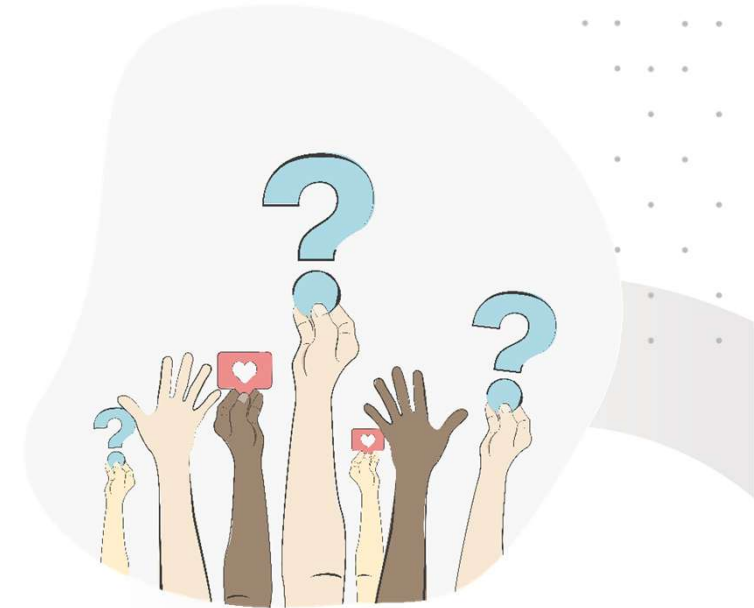
Data stewardship and literacy

Collaboration and stakeholder buy-in

Data Strategy and Executive support

What else can you remember / come up with?

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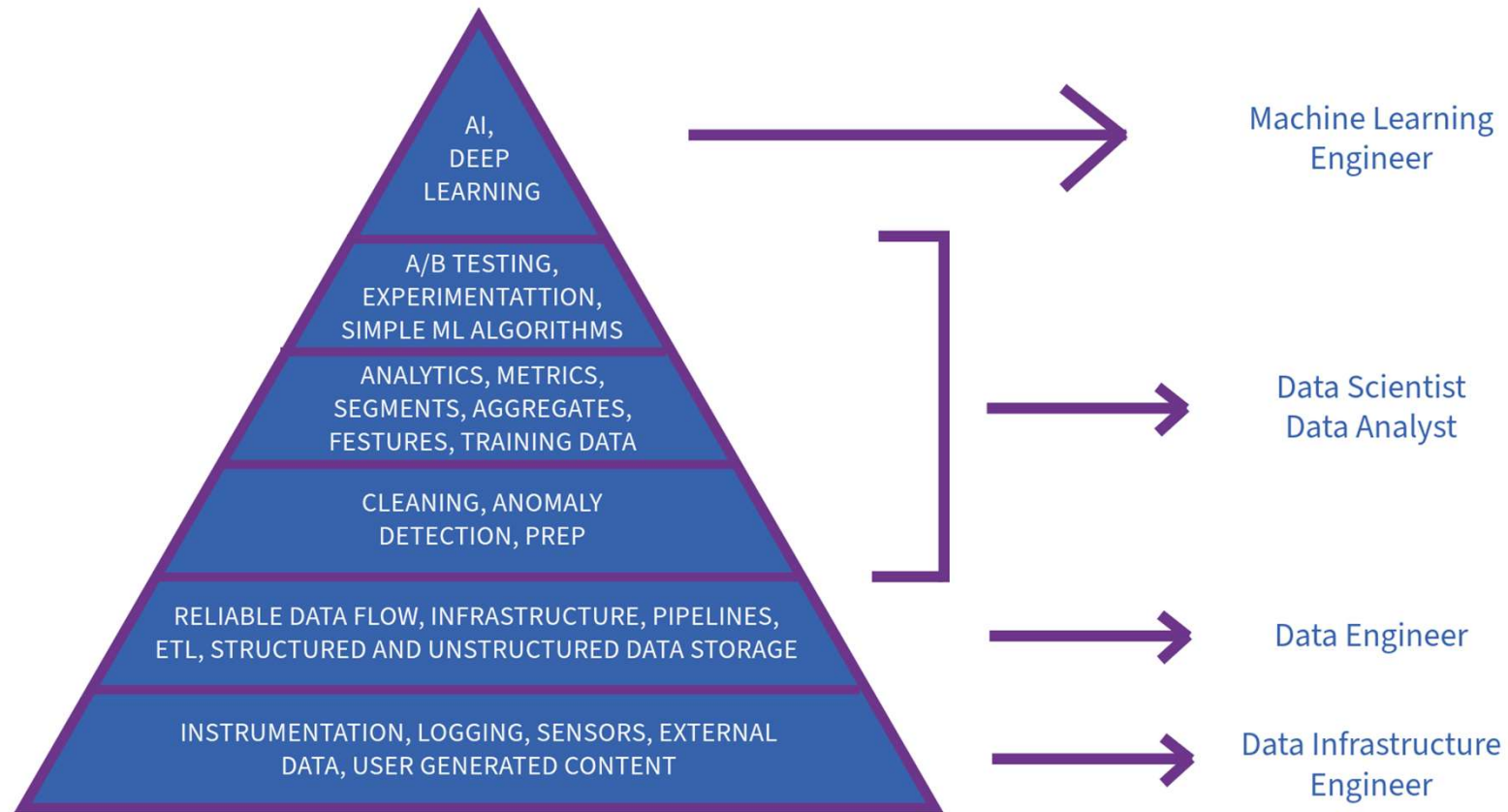
**Submit your responses to
the chat!**



Building a Data-Driven Culture



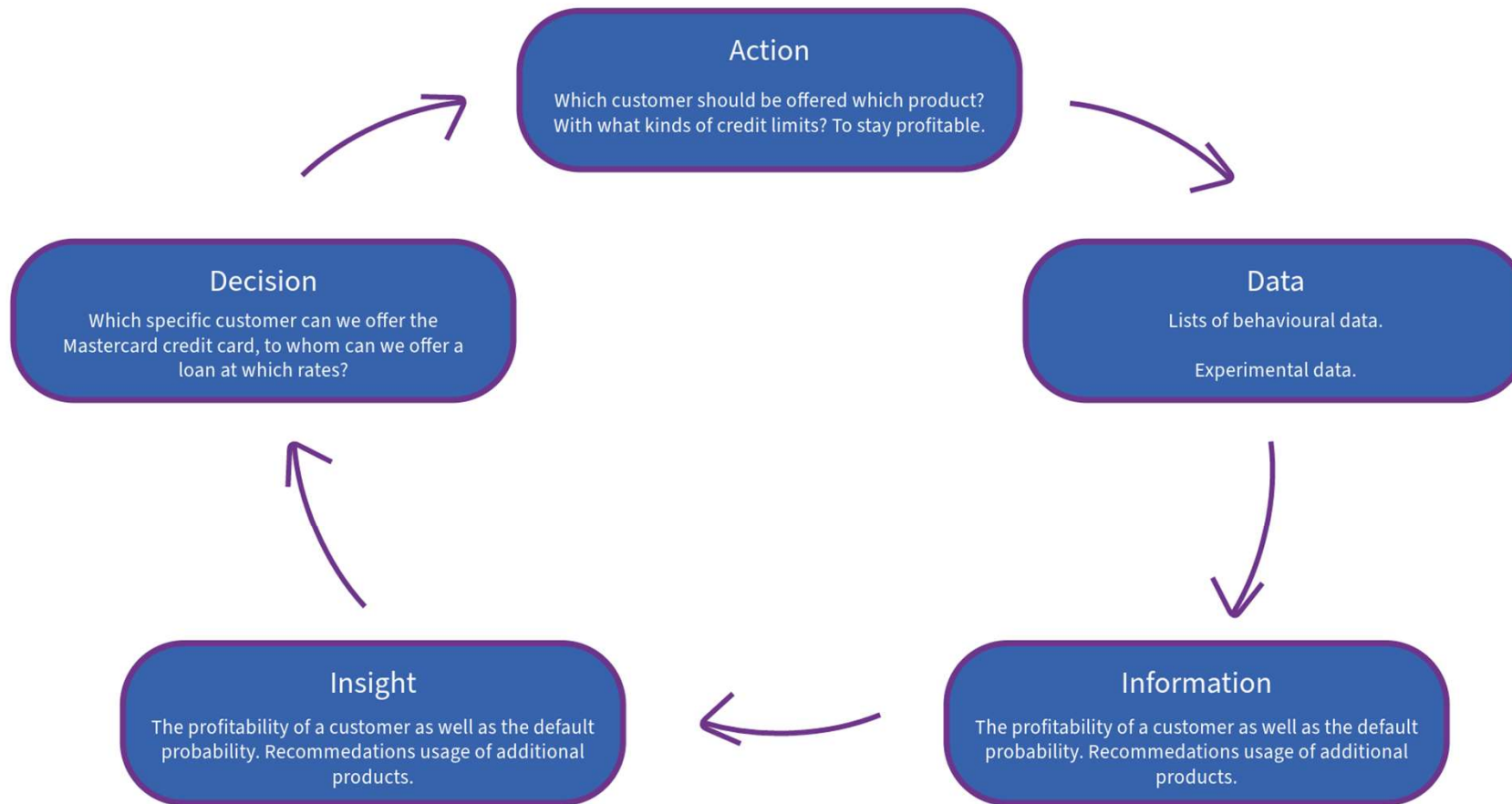
The Data Science Hierarchy of Needs



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Turning Data into Actionable Insight

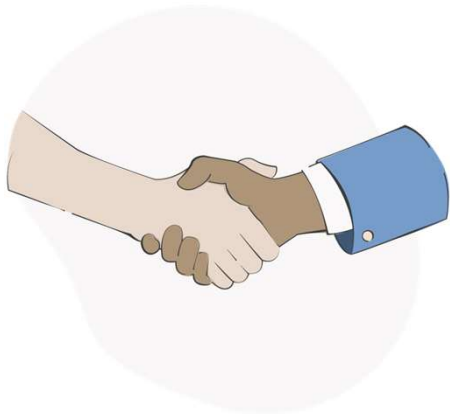


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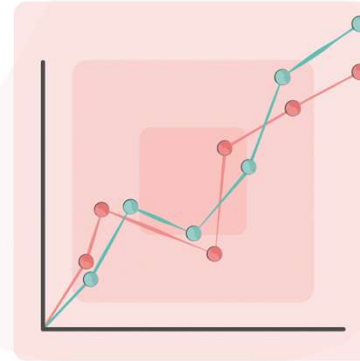


First Steps

What strategies should Credit Bank use to generate a data-driven culture?



Build relationships



Choose transparency in algorithms



Celebrate and embrace small wins



Raise data literacy

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Learning journal - reminder

Your tutor will remind you about the importance of your learning journal and point you to the section in the Handbook that discusses it.

You will be also shown where the post-webinar activity is on the Hub, which will help you set up your first learning journal on GitHub.

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Transitioning from Small to Big Data

Why does this happen?

Normally, we'd expect the transition from small to big data to happen because of the following:

Overwhelming Data

Performance Bottlenecks

Missed Opportunities



Big Data

Defining the Extremes

Can you identify small and big data?

Data Examples A

Personal fitness tracker recording daily steps and calories burned

A collection of contacts in a mobile phone's address book

A simple spreadsheet tracking monthly expenses



Small Data: like a crystal-clear lake

Data Examples B

Social media platforms analysing billions of user interactions daily

Weather monitoring stations collecting vast amounts of climate data worldwide

Online retailers tracking millions of transactions and customer behaviors



Big Data: like an ocean

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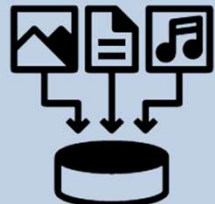
The Characteristics of Data

The 5 Vs of data

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Volume



Variety



Velocity



Veracity



Value

The 5 Vs of data framework

Raise data literacy



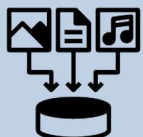
Small Data vs Big Data

In the context of **value...**

Big Data	Small Data
Big data offers valuable insights and business value through advanced analytics	Small data delivers immediate value through straightforward analysis
It uncovers hidden patterns, trends, and correlations	It addresses specific operational questions efficiently
This enables predictive modeling and data-driven decision-making	Small data supports day-to-day decision-making processes effectively
Big data drives innovation and competitive advantage	Small data supports day-to-day decision-making processes effectively



Volume



Variety



Velocity



Veracity



Value



Case study: Credit Bank Corporation's data analytics dashboard

Knowledge Check Poll

Which of the following best represents the "veracity" characteristic of big data in the context of Credit Bank Corporation's HR analytics dashboard project?

- A) The speed at which employee data is generated from different HR systems
- B) The accuracy and trustworthiness of the employee data collected
- C) The diverse range of data sources used
- D) The large size of the employee data being used

Feedback: B - The accuracy and trustworthiness of the employee data collected.

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Fundamentals of Data

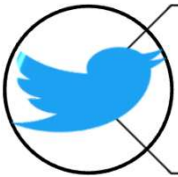


Data Deluge

Appreciating data sizes

As we live in an increasingly data-driven world, it's important to understand how we quantify the vast amounts of data being generated and stored.

Modern data is growing rapidly, every minute:



500 thousand tweets
are sent

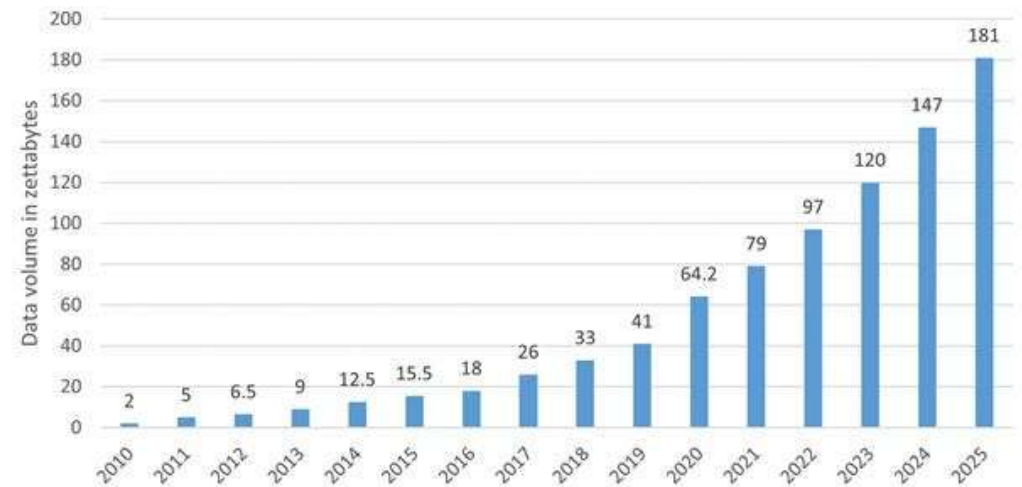


500 million instant
messages are sent



5 terabytes of data is
posted on Facebook

Volume of data created and replicated worldwide (source: IDC)

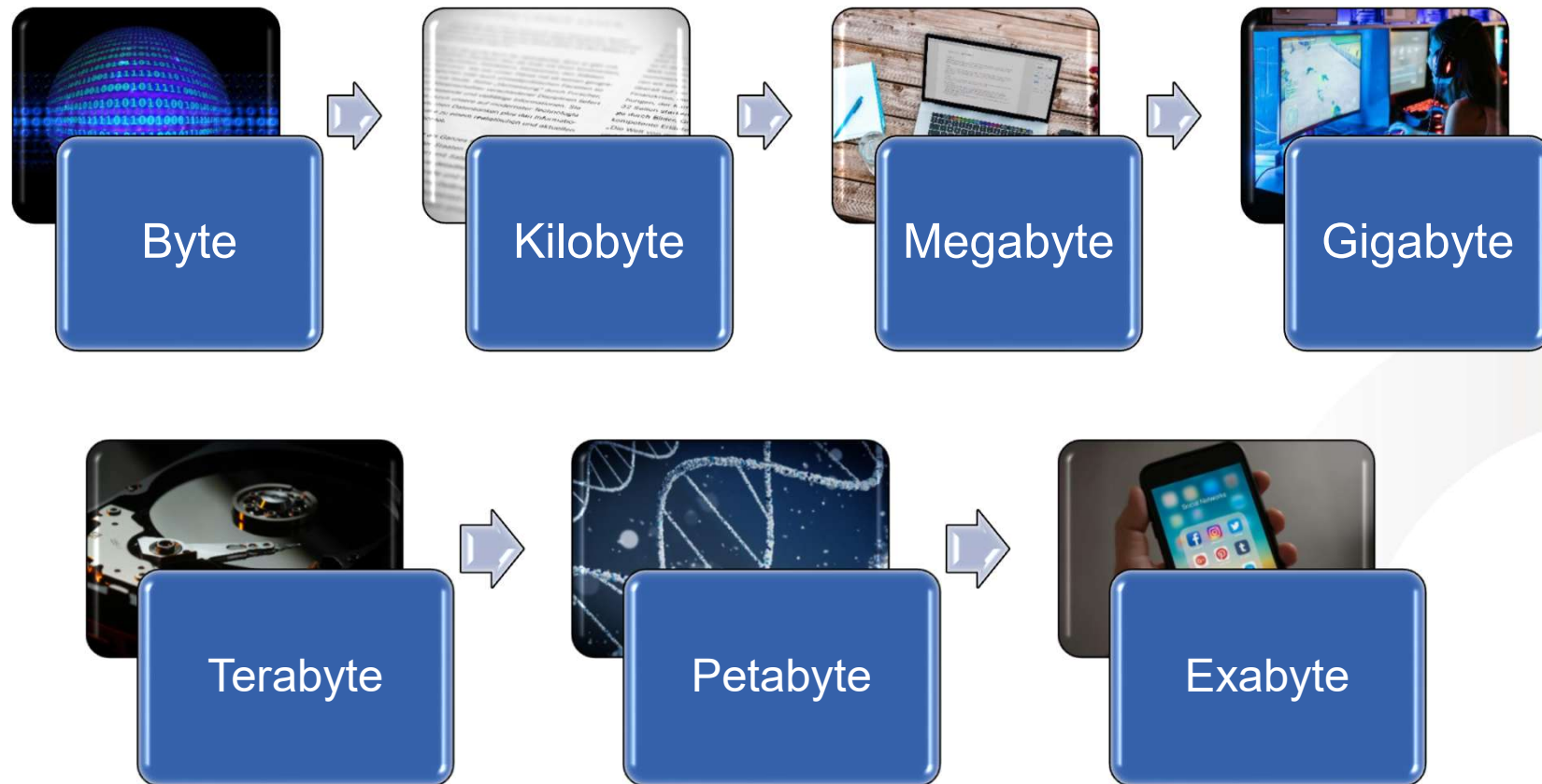


Modern data is growing rapidly!

Image source: Medium.com

Appreciating Data Sizes

From Byte to Exabyte...



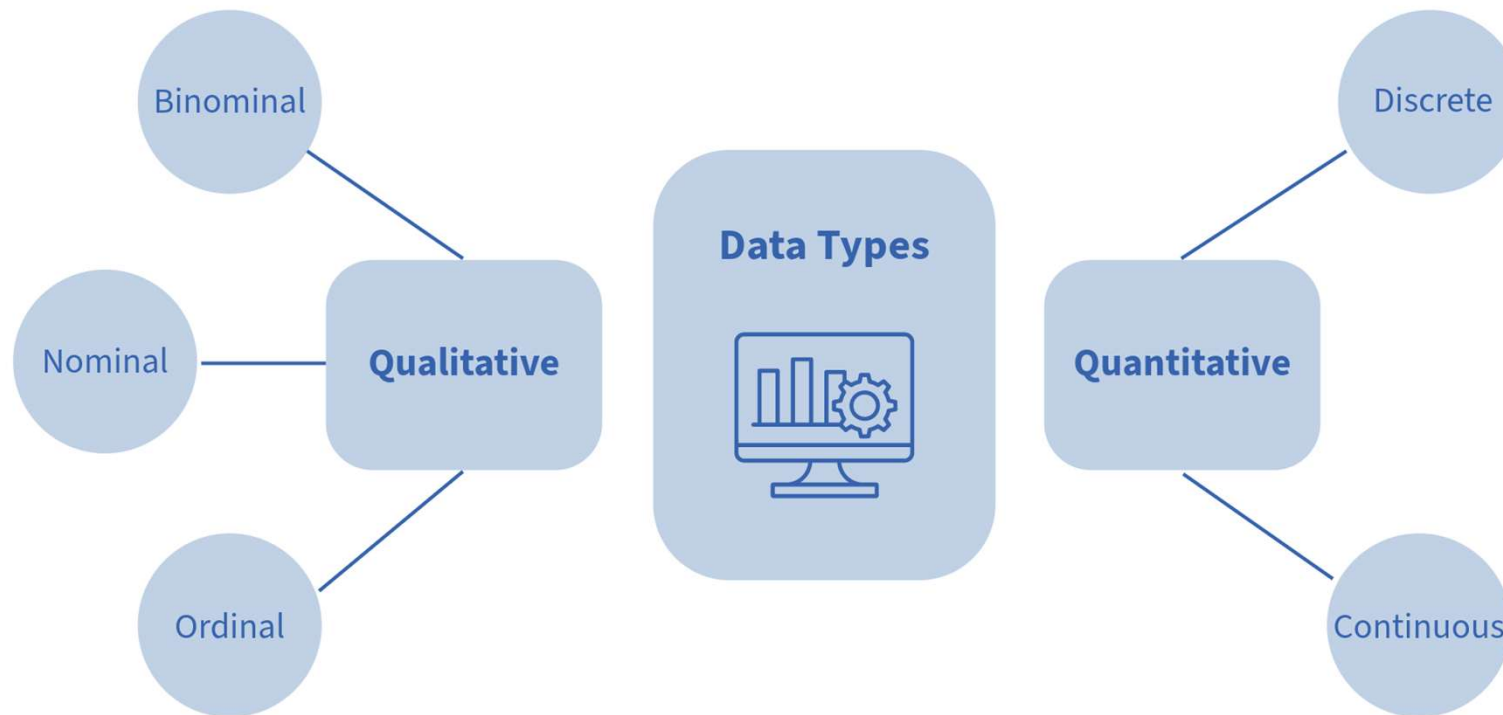
Data sizes and everyday examples from Byte to Exabyte

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The Different Types of Data

A hierarchy of two main groups...



The Hierarchy of Qualitative and Quantitative Data Types

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Knowledge Check Poll

If a dataset containing employee performance reviews from the past 10 years is approximately 50 Gigabytes (GB) in size...

What would be the next larger data size unit that could represent this dataset?

- A) Megabyte (MB)
- B) Terabyte (TB)
- C) Petabyte (PB)
- D) Exabyte (EB)

Feedback: B – Terabyte (TB)

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the chat!**



Knowledge Check Poll

Which type of data would be most suitable for tracking employee attendance in Credit Bank Corporation's HR analytics dashboard?

- A) Qualitative data (nominal)
- B) Qualitative data (ordinal)
- C) Quantitative data (discrete)
- D) Quantitative data (continuous)

Feedback: A, or C, or D (context matters!)

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Identifying Standards and Engineering Best Practice



Data Formats

JSON, CSV and XML...



Social Media



Financial
Institutions



Healthcare

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Cloud Computing Standards

And the impact on business...

As cloud adoption accelerates, standards and best practices ensure consistent, reliable, and secure cloud implementations.

We will now explore two prominent cloud computing standards:

AWS Well-Architected Framework

OpenAPI



The impact of API standards example:

Healthcare



Regulatory Requirements

Privacy, security and compliance...



*General Data Protection Regulations
(GDPR)*



*Information Security Management
System (ISO) 27001*

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Engineering Best Practices

Vital for developing robust, scalable, and dependable systems...



Scalability



Engineering Best Practices



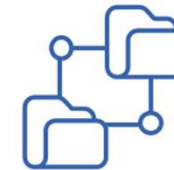
Reliability



Security



Performance Optimisation



Data Documentation

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Data Stewardship Principles

Quality, governance and ethics

As data continues to proliferate exponentially, establishing a robust framework for data stewardship becomes imperative.

There are three fundamental principles of responsible and effective data stewardship, as follows:



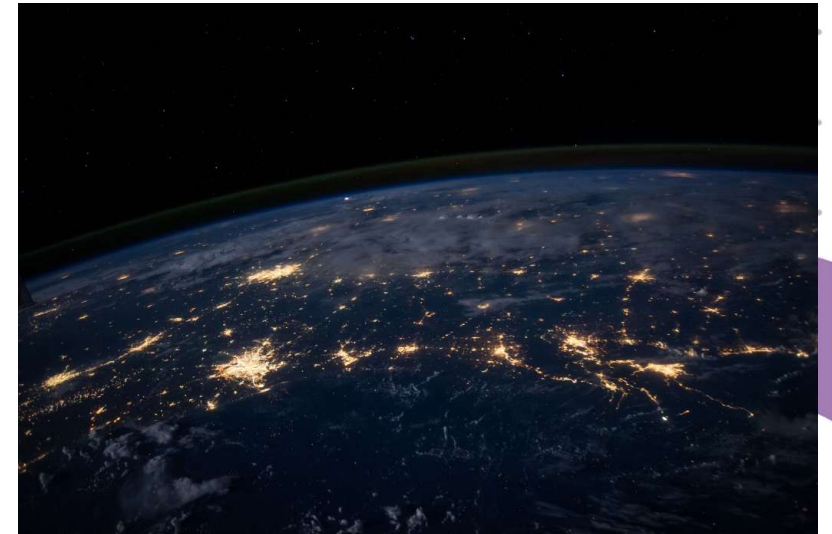
Data quality



Data governance



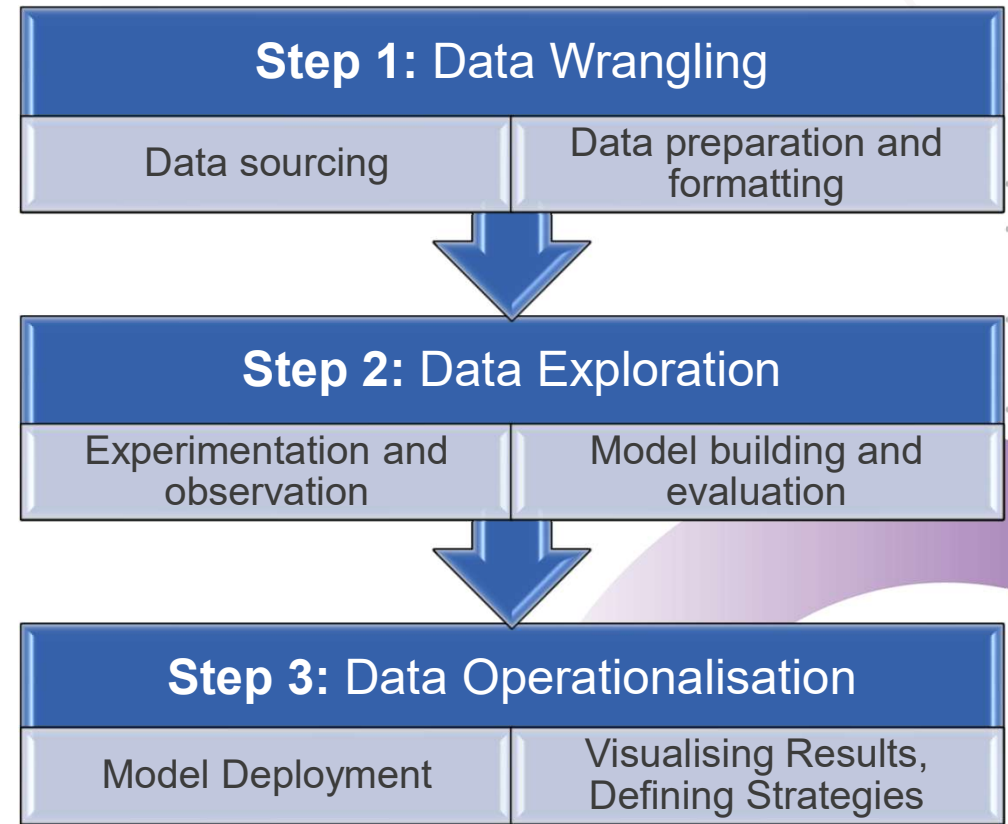
Data ethics



Adding Value With Data

The three stages of adding value with data

- The tangible results (deliverables) of adding value with data are Data Products and Services
- Data Engineering is primarily active during Data Wrangling and Data Operationalisation
- Data Science and Data Analytics are mainly concerned with Data Exploration



The three stages of adding value with data

Data Teams

Who do they include?



Software Engineer



Data Teams



Chief Data Officer



Data Scientist



Data Engineer



Data Analyst

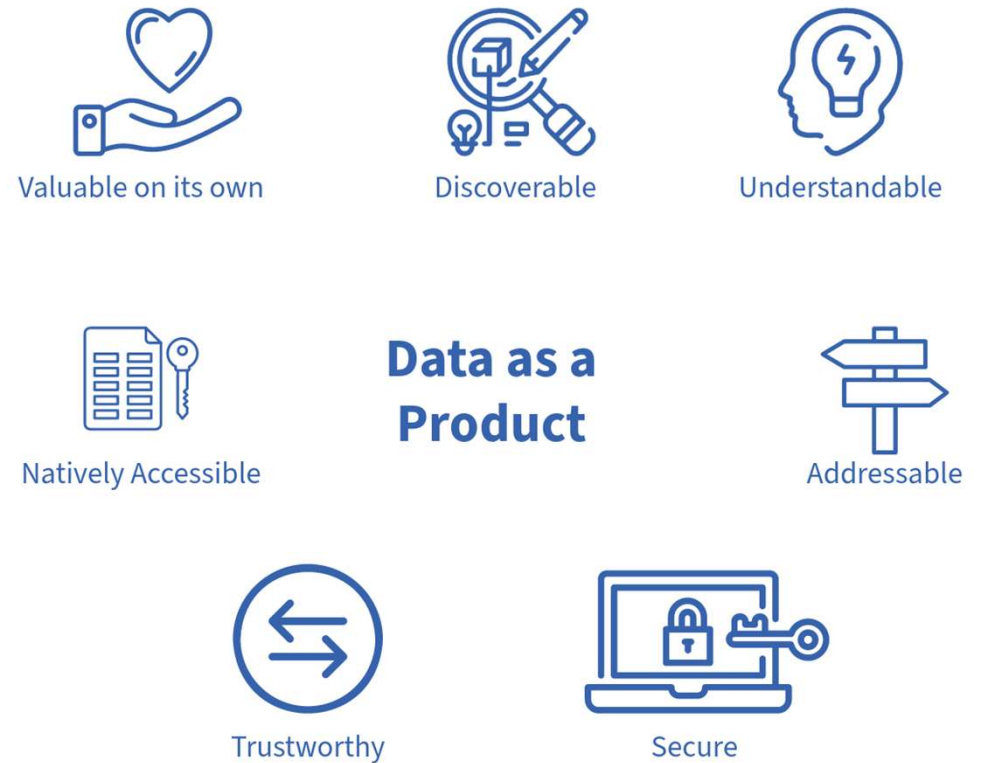
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Data as a Product

A product in itself and by itself...

- Data can be monetised directly or indirectly
- Data brokers collect and sell aggregated data for targeted advertising
- Companies like Google and Facebook monetise user data through advertising
- Data engineers transform raw data into valuable products
- They ensure data is valuable, accessible, trustworthy, discoverable, and interoperable



Staff Retention Example

Will Peter stay with his employer...?

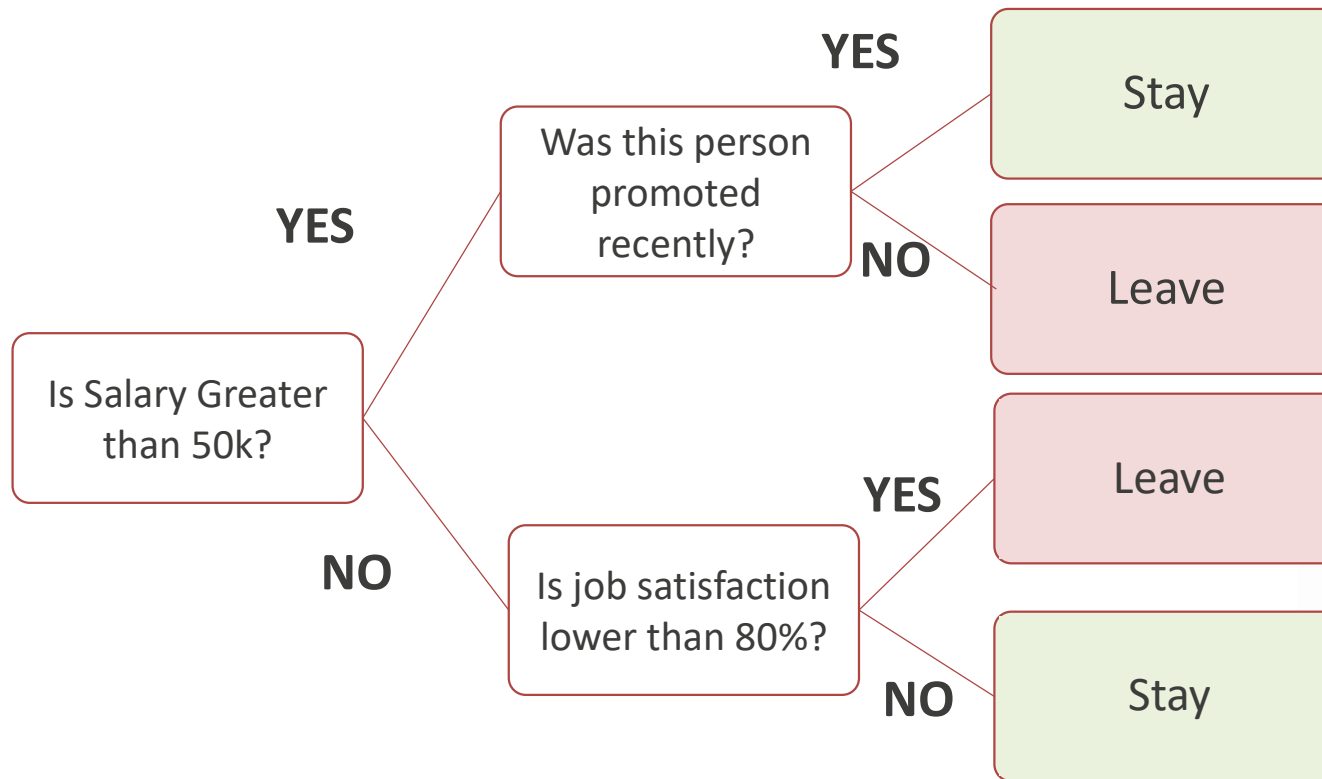
Datapoint	Value
Name:	Peter
Satisfaction Level:	0.80
Last Evaluation Score:	0.86
Number of Projects:	5
Average Monthly Hours	262
Time with Company (Years)	6
Work Accidents	0
Promotions in Last 5 Years	0
Department	Sales
Salary	£45,000
Did he resign?	?



Decision Trees

An introduction...

Decision trees are machine learning algorithms that represent decision-making processes in a flowchart-like structure, facilitating clear and interpretable understanding of outcomes.



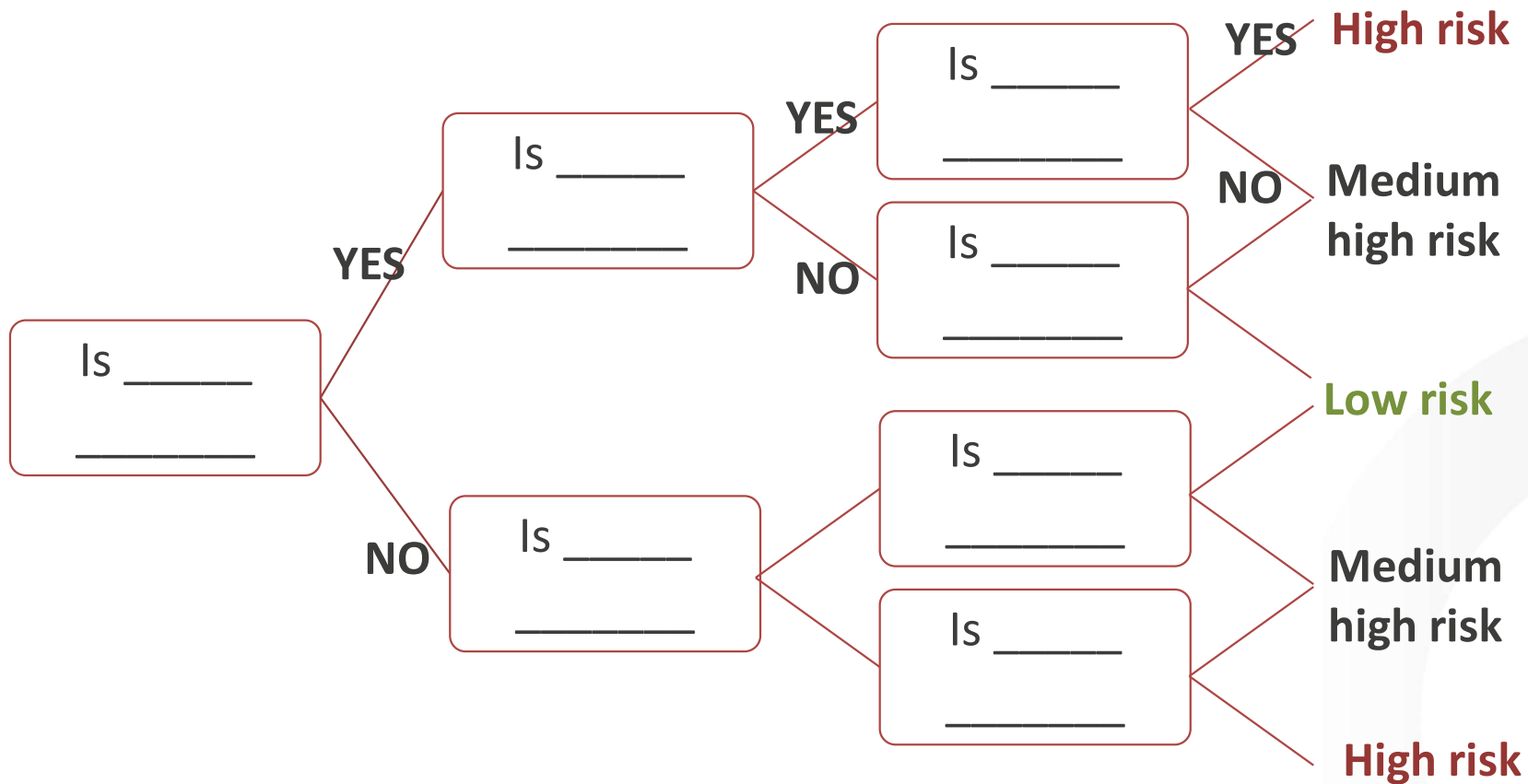
An example decision tree

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Decision Trees

The deep decision tree model...



An example deep decision tree model

Building a Decision Tree

Group discussion

Building an Employee Churn Decision Tree

Objective: To illustrate how actionable insights are generated from data.

Brief: Working in small groups, identify the most useful factors for HR employee churn. Try to build a decision tree that classifies employees into risk groups based on these factors.

Final output:

- By completing this exercise you should produce a decision tree that can be tested on new (unseen) data to deliver insights
- You should also be able to develop a strategy of next steps based on the decision tree

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Group discussion

Integrating Diverse Data Sources

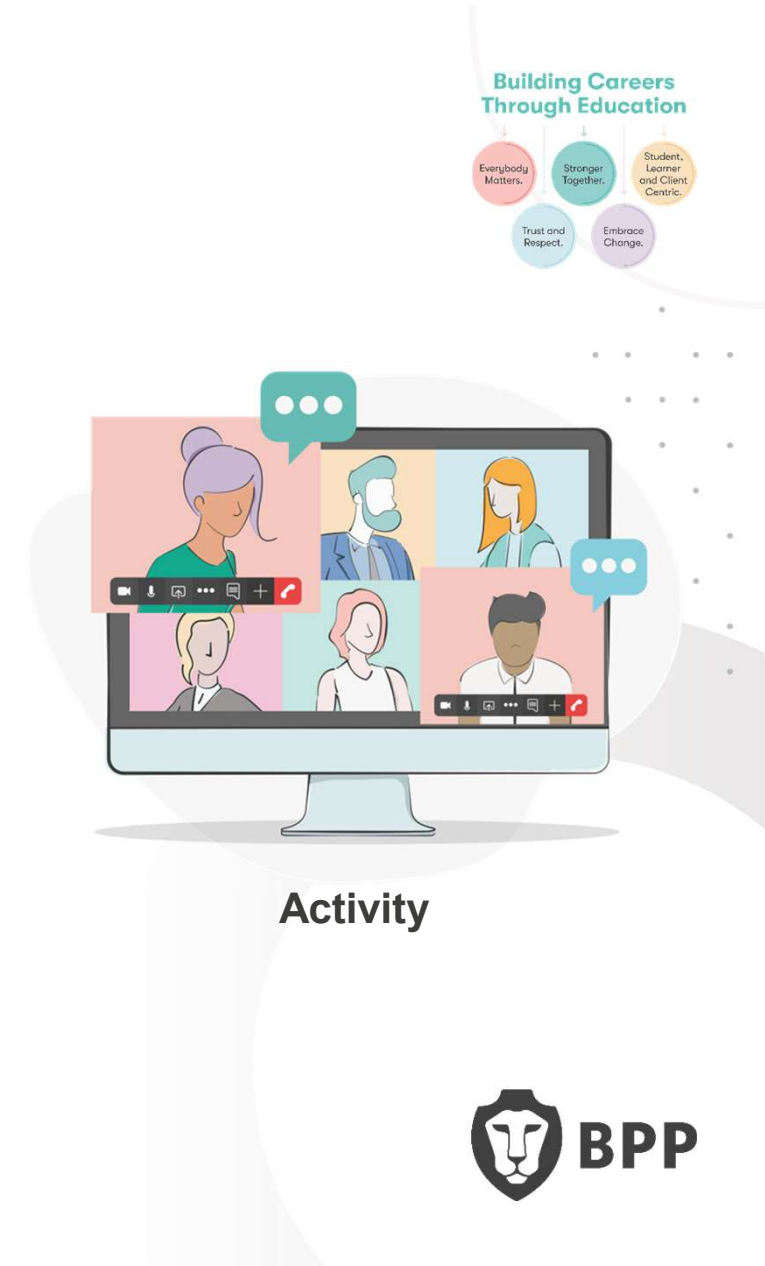
Activity

We must now imagine we are part of a team of data engineers within Credit Bank Corporation tasked with integrating employee performance data from various sources into the HR analytics dashboard. The data sources include:

- Employee performance reviews (CSV file)
- Learning Management System (LMS) data (SQL database)
- Employee engagement survey responses (JSON file)

Instructions:

1. Collect data from provided sources
2. Clean and preprocess data for quality and consistency
3. Transform data into suitable format for analysis
4. Integrate transformed data into unified dataset



Key Learning Summary

The key takeaways from this session are as follows:

- Building a data-driven culture involves strategies like building relationships, choosing transparency in algorithms, celebrating small wins, and raising data literacy
- The 5 Vs of Big Data are Volume, Variety, Velocity, Veracity, and Value
- Appreciating different data size units, from bytes to exabytes, is crucial for quantifying modern data volumes
- Cloud computing standards like the AWS Well-Architected Framework ensure consistent, reliable, and secure cloud implementations



Key Learning Summary (Cont'd)

The key takeaways from this session are as follows:

- Regulatory requirements like GDPR and ISO 27001 govern data privacy, security, and compliance
- Data stewardship principles include data quality, data governance, and data ethics
- Adding value with data involves three stages: data wrangling, data exploration, and data operationalisation
- Data can be monetised directly or indirectly, treated as a product itself
- Decision trees are machine learning models that represent decision-making processes in a flowchart structure, facilitating interpretable understanding of outcomes





Thank you

**Do you have any questions,
comments, or feedback?**

